# An explicit nonreversible Cauchy semigroup on the circle: generator, entropy and dual boundaries

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Reviewed editorial draft, October 30, 2026

**Abstract**

A fractional circle generator with sine drift admits an explicit transition kernel obtained by harmonic extension along a rational disk flow. We construct its conservative Feller semigroup and invariant wrapped Cauchy law, identify the full generator graph on continuous functions and its Laplace resolvent, and prove graph density of time-smoothed trigonometric data. For every initial probability law the positive-time forward density has finite relative entropy. Its exact entropy production is a singular jump-Bregman integral, even though the generator is nonsymmetric under its invariant law. An elementary nonsharp modified logarithmic-Sobolev bound yields exponential decay from initial time, with infinite initial entropy allowed. The proof avoids assigning an unproved backward-generator domain to a logarithmic density. The construction belongs to classical harmonic and semigroup theory and is related to known Cauchy-driven phase dynamics. Exact statements are mapped to a frozen Lean development. Exact single-harmonic corrections and real-polynomial tests distinguish the special sine drift from higher harmonics, Haar energies and diffusion chain rules. We also prove actual weak forward-law evolution, point-mass dual norm obstructions and the exact regularity of time-averaged law functionals. The drift-free benchmark has a complete maximal Haar-$L^2$ fractional generator domain and unsmoothed finite Fourier graph core. A zero-drift Fourier test separates a bounded transposed-generator value from norm differentiability of an averaged dual orbit. Novelty of the explicit combined package is unresolved.

## 1 The circle flow and the conclusions

Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ and let $m$ be normalized Haar measure, so in angle coordinates $dm = d\theta/(2\pi)$. For $a \in \mathbb{D}$, the Poisson kernel is

$$p_a(w) = \frac{1 - |a|^2}{|w - a|^2}, \qquad w \in \mathbb{T}.$$

It is a positive probability density with respect to $m$. Fix a real parameter $|q| < 1$, and put

$$r = \frac{2q}{1 - q^2}, \qquad \alpha = \frac{1 + q^2}{1 - q^2}, \qquad c_q = \frac{1 - |q|}{2(1 + |q|)}. \tag{1}$$

For $0 \le \lambda \le 1$ define

$$F_\lambda(z) = \frac{q(1 - \lambda) + (q^2 + \lambda)z}{1 + q^2\lambda + q(1 - \lambda)z}, \qquad \phi_t = F_{e^{-\alpha t}} \quad (t \ge 0). \tag{2}$$

For $t > 0$ define the transition probability

$$K_t(z, dw) = p_{\phi_t(z)}(w)\, m(dw), \qquad K_0(z, \cdot) = \delta_z, \tag{3}$$

and $P_t f(z) = \int f(w) K_t(z, dw)$. Our invariant reference law is $\nu_q(dw) = p_q(w) m(dw)$.

**Theorem 1.1.** *The kernels* (3) *form a conservative Markov semigroup. On* $C(\mathbb{T};\mathbb{C})$, $P_t$ *is a strongly continuous positive contraction semigroup with invariant probability* $\nu_q$. *Its closed generator* $A$ *acts on trigonometric polynomials as*

$$A = -|D_\theta| - r\sin\theta\,\partial_\theta, \qquad |D_\theta|e^{in\theta} = |n|e^{in\theta}. \tag{4}$$

*Its full graph is characterized by*

$$(f,g)\in \operatorname{graph} A \quad\Longleftrightarrow\quad P_t f - f = \int_0^t P_s g\,ds \quad \text{for every } t\ge 0. \tag{5}$$

*Time averages of trigonometric polynomials are graph dense. For* $\beta>0$ *the positive Laplace resolvent* $R_\beta f = \int_0^\infty e^{-\beta t}P_t f\,dt$ *is the inverse of* $\beta - A$, *with norm at most* $1/\beta$.

*For every Borel probability measure* $\mu$, *write* $\mu_t = \mu K_t$. *At every* $t>0$ *it has a positive continuous density* $f_t = d\mu_t/d\nu_q$, *and*

$$\frac{d}{dt}D(\mu_t\|\nu_q) = -\int_{\mathbb{T}^2} \frac{2\,B(f_t(z),f_t(w))}{|w-z|^2 p_q(w)}\,\nu_q(dz)\nu_q(dw), \tag{6}$$

*where* $B(a,b) = a\log(a/b) - a + b$ *and the diagonal integrand is defined to be zero. The integral is finite, and*

$$D(\mu_t\|\nu_q) \le e^{-c_q t}D(\mu\|\nu_q), \qquad t\ge 0, \tag{7}$$

*in the extended nonnegative reals. The invariant law is unique. For* $q\neq 0$ *the generator is not Hermitian even on its finite Fourier domain in* $L^2(\nu_q)$.

Here $D(\mu\|\nu) = \int f\log f\,d\nu$ when $\mu = f\nu$ and this integral is finite, with the usual extended-value definition otherwise. The constant $c_q$ is a convenient lower bound, not a sharp entropy constant. The generator graph is on continuous functions with the supremum norm. We will not identify it with an unproved Sobolev domain or replace the smoothed graph core by an unsmoothed-core assertion.

## 2 Disk geometry and harmonic transition kernels

The denominator in (2) has modulus at least $1+q^2\lambda - |q|(1-\lambda) > 0$ on the closed disk. Writing its numerator and denominator as $N$ and $H$, direct multiplication gives

$$\begin{aligned} |H|^2 - |N|^2 = {}& (1-\lambda)^2(1-q^2)|1+qz|^2 \\ & + \lambda^2(1+q^2)^2(1-|z|^2) \\ & + 2\lambda(1-\lambda)(1+q^2)(1-q^2|z|^2). \end{aligned} \tag{8}$$

All terms are nonnegative on $\overline{\mathbb{D}}$, and the first is strictly positive for $\lambda<1$. Thus $F_\lambda$ maps the closed disk strictly into $\mathbb{D}$ when $\lambda<1$, and $F_1(z)=z$. Each map is holomorphic on a neighborhood of the closed disk.

The matrix defining this fractional-linear map is

$$M_\lambda = \begin{pmatrix} q^2+\lambda & q(1-\lambda) \\ q(1-\lambda) & 1+q^2\lambda \end{pmatrix}.$$

Its eigenvectors $(q,1)$ and $(1,-q)$ have eigenvalues $1+q^2$ and $\lambda(1+q^2)$, respectively. Hence $M_\lambda M_\eta = (1+q^2)M_{\lambda\eta}$, proving

$$\phi_{s+t} = \phi_s\circ\phi_t, \qquad \phi_0(z)=z, \qquad \phi_t(q)=q. \tag{9}$$

Differentiating (2) at time zero gives

$$\partial_t \phi_t(z)|_{t=0} = V(z) := -z + \frac{q}{1-q^2}(1-z^2). \tag{10}$$

The denominator lower bound makes this differentiation uniform on the closed disk. It also gives joint continuity in $t, z$ and uniform convergence $\phi_t(z) \to z$ as $t \downarrow 0$.

For $f \in C(\mathbb{T}; \mathbb{C})$, let $u_f$ be its harmonic Poisson extension to the disk, continuous up to the boundary. Then

$$P_t f(z) = u_f(\phi_t(z)).$$

The function $u_f \circ \phi_s$ is harmonic in $\mathbb{D}$ and continuous on its closure. Its boundary values are $P_s f$. The uniqueness of the Dirichlet problem therefore gives $u_{P_s f} = u_f \circ \phi_s$, whence

$$P_t P_s f = u_f \circ \phi_s \circ \phi_t = P_{s+t} f.$$

Equality on continuous functions determines finite Borel measures on the compact circle. This proves the exact Chapman–Kolmogorov identity for (3), not just agreement of a formal generator.

The Poisson kernel is nonnegative with integral one, so $P_t$ is positive, preserves constants, and is a supremum-norm contraction. Continuity of $u_f \circ \phi_t$ proves the Feller property. For strong continuity at zero, use uniform continuity of $u_f$ on $\overline{\mathbb{D}}$ and uniform convergence of $\phi_t$ to the identity. One can also verify boundary continuity of the Poisson extension directly: split its integral into a small arc where $f$ oscillates little and its complement, where $p_a(w) \le (1-|a|^2)/(\varepsilon - |a-z|)^2$ tends uniformly to zero as $a \to z \in \mathbb{T}$.

Finally $\int_{\mathbb{T}} f \, d\nu_q = u_f(q)$, so

$$\int_{\mathbb{T}} P_t f \, d\nu_q = u_{P_t f}(q) = u_f(\phi_t(q)) = u_f(q).$$

This proves invariance. The kernels are Borel measurable in their initial point because their positive-time densities are jointly continuous, and at zero they are the measurable identity kernel.

## 3 The closed generator and the precise graph core

For $n \ge 0$, the Poisson moment identity gives

$$P_t(z^n) = \phi_t(z)^n, \qquad P_t(\bar{z}^n) = \overline{\phi_t(z)}^{\,n}. \tag{11}$$

Uniform differentiation using (10) shows that these modes belong to the continuous-function generator domain. For $n \ge 1$,

$$Az^n = -nz^n + \frac{nr}{2}(z^{n-1} - z^{n+1}),$$

with the conjugate formula for negative frequencies and $A1 = 0$. Since $\partial_\theta z^n = inz^n$, these are exactly (4). This statement concerns actual supremum-norm difference quotients.

Define $f \in D(A)$ when $h^{-1}(P_h f - f)$ has a supremum-norm limit $g$ as $h \downarrow 0$, and set $Af = g$. The limit is unique.

**Proposition 3.1.** *The equivalence* (5) *holds, and its right-hand side defines a closed linear graph.*

*Proof.* If $Af = g$, the semigroup identity gives the right derivative $\partial_t^+ P_t f = P_t g$. For $t > 0$ the left difference quotient is $P_{t-h}(P_h f - f)/h$, also tending to $P_t g$ by contraction and strong continuity. Thus the orbit is differentiable for $t > 0$ with continuous derivative $P_t g$, and the vector-valued fundamental theorem of calculus gives (5).

Conversely, divide that identity by $t$ and let $t \downarrow 0$. Strong continuity gives $t^{-1} \int_0^t P_s g \, ds \to g$, so $f \in D(A)$ and $Af = g$. For a convergent sequence of graph pairs $(f_n, g_n)$, contraction bounds the difference of the integrals by $t\|g_n - g\|_\infty$. Pass to the limit in (5); the limiting pair is again in the graph. Linearity follows from the same identity. □

For $h > 0$ define the time average

$$J_h f = \frac{1}{h} \int_0^h P_s f \, ds. \tag{12}$$

A shift of the integration interval gives

$$J_h f \in D(A), \qquad A J_h f = (P_h f - f)/h. \tag{13}$$

Indeed $(P_t J_h f - J_h f)/t$ is $1/h$ times the difference of the averages of $P_s f$ on $[h, h+t]$ and $[0, t]$. Letting $t \downarrow 0$ proves the formula. Since $J_h f \to f$, the generator domain is dense. If $f \in D(A)$, (5) also gives $A J_h f = J_h A f \to Af$, so these averages approximate in the full graph norm.

Let $\mathcal{P}$ be the real-and-complex trigonometric polynomials. They are uniformly dense in $C(\mathbb{T}; \mathbb{C})$. For fixed $h > 0$, the map

$$f \longmapsto (J_h f, (P_h f - f)/h)$$

is continuous into the product of supremum-norm spaces: its two component norms are bounded by 1 and $2/h$. For a graph pair $(f, Af)$, first choose $h$ small enough that its averaged pair is close; then choose $p \in \mathcal{P}$ close enough to $f$ that the corresponding averaged pair is close. Consequently

$$\overline{\{(J_h p, A J_h p) : h > 0, \ p \in \mathcal{P}\}} = \text{graph } A. \tag{14}$$

This is the precise smoothed core assertion. It does not identify the closure of the unsmoothed polynomial graph.

## 4 Laplace resolvent on the full domain

For $\beta > 0$, contraction makes the Bochner integral

$$R_\beta f = \int_0^\infty e^{-\beta t} P_t f \, dt$$

absolutely convergent, with norm at most $\|f\|_\infty/\beta$. It is positive on real nonnegative inputs. For a finite integral $u_T = \int_0^T e^{-\beta t} P_t f \, dt$, the shifted-orbit identity is

$$P_h u_T = e^{\beta h} \int_h^{T+h} e^{-\beta s} P_s f \, ds.$$

Differentiate at $h = 0$ from the right to obtain

$$u_T \in D(A), \qquad A u_T = \beta u_T + e^{-\beta T} P_T f - f. \tag{15}$$

The tails of both sides converge in supremum norm. Closedness of the graph gives

$$R_\beta f \in D(A), \qquad (\beta - A) R_\beta f = f. \tag{16}$$

For $u \in D(A)$, the orbit derivative from Proposition 3.1 yields

$$\frac{d}{dt}(e^{-\beta t}P_t u) = -e^{-\beta t}P_t(\beta u - Au).$$

Integrate from zero to infinity; the endpoint at infinity vanishes by contraction. Hence $R_\beta(\beta u - Au) = u$ as well. This proves the full-domain inverse formula and the bound

$$\beta\|u\|_\infty \le \|\beta u - Au\|_\infty.$$

No self-adjointness was used. The next calculation shows why it would be inappropriate to insert it.

## 5 An exact nonsymmetry witness

Under $\nu_q$, the Poisson moments satisfy $\int z^n \, d\nu_q = q^n$ for $n \ge 0$. Use $f(z) = z$ and $g(z) = z^2$, both in the finite Fourier generator domain. Since $q$ and $r$ are real,

$$\begin{aligned}\langle f, Ag\rangle_{L^2(\nu_q)} &= -2q + r(1 - q^2) = 0,\\ \langle Af, g\rangle_{L^2(\nu_q)} &= -q + \tfrac{r}{2}(q^2 - 1) = -2q.\end{aligned}$$

Here the inner product is conjugate linear in its first argument. If $q \neq 0$ the pairings differ. Thus no Hermitian realization agreeing with this core can be inferred from the invariant Poisson law. Invariance, positivity and entropy decay remain valid; they do not require reversibility.

## 6 Forward densities and a stationary Bregman identity

For an arbitrary probability measure $\mu$, Fubini gives the positive-time density relative to $\nu_q$:

$$f_t(w) = \int_{\mathbb{T}} \frac{p_{\phi_t(z)}(w)}{p_q(w)} \, \mu(dz). \tag{17}$$

The ratio is jointly continuous and strictly positive on the compact product circle for fixed $t > 0$. It has a positive minimum and finite maximum, independent of $\mu$. On every compact time interval inside $(0, \infty)$, the denominators stay uniformly away from zero. Ordinary differentiation of the rational kernel, uniformly in $z, w$, proves that $t \mapsto f_t$ is continuously differentiable in supremum norm. The same denominator bound gives a bounded angular derivative in $w$, hence a Lipschitz bound in chord distance on $\mathbb{T}$.

In particular $\int f_t \, d\nu_q = 1$ and $E(t) = \int f_t \log f_t \, d\nu_q$ is finite and differentiable for $t > 0$. Positivity and the uniform bounds on compact time intervals justify the usual scalar derivative,

$$E'(t) = \int_{\mathbb{T}} \dot{f}_t \log f_t \, d\nu_q, \tag{18}$$

since $\int \dot{f}_t \, d\nu_q = 0$. This is a statement about actual forward laws and their Radon–Nikodym density, not an assumed backward-generator domain for $\log f_t$.

For positive $a, b$, set $B(a, b) = a\log(a/b) - a + b$. It is nonnegative by convexity of $s \log s - s$. If $f\nu_q$ is a probability law with positive continuous $f$, invariance and the two marginals of $\nu_q(dz)K_h(z, dw)$ give the exact identity

$$\begin{aligned}&\int_{\mathbb{T}^2} B(f(z), f(w)) \, \nu_q(dz)K_h(z, dw)\\ &\qquad = \int_{\mathbb{T}} f \log f \, d\nu_q - \int_{\mathbb{T}} \log f \, d((f\nu_q)K_h).\end{aligned} \tag{19}$$

Indeed expansion of $B$ gives four terms. The first integrates to $\int f \log f \, d\nu_q$; the mixed term is the last term in (19); and the two linear terms cancel because both stationary marginals are $\nu_q$. All integrands are bounded, so Fubini is immediate. For $f = f_t$, Chapman–Kolmogorov identifies the last law as $\mu_{t+h}$. Divide by $h$ and use the supremum-norm derivative of $f_t$ to conclude

$$\lim_{h \downarrow 0} \frac{1}{h} \int B(f_t(z), f_t(w)) \, \nu_q(dz) K_h(z, dw) = -E'(t). \tag{20}$$

# 7 The singular jump limit and its domination

Let $v_h(z) = \phi_h(z)$, $z \in \mathbb{T}$. Equation (10) gives

$$\frac{d}{dh}(1 - |v_h(z)|^2)\Big|_{h=0} = 2. \tag{21}$$

To check the independence of $q$, compute $-2\,\mathrm{Re}(\bar{z}V(z))$. The $-z$ term contributes two, while $\bar{z}(1 - z^2) = \bar{z} - z$ is purely imaginary. The derivatives are uniform on the compact circle. For $z \neq w$ it follows that

$$\frac{p_{v_h(z)}(w)}{h} \longrightarrow \frac{2}{|w - z|^2}. \tag{22}$$

A pointwise limit alone does not justify the singular integral in (6); we give a bound that also handles points arbitrarily close to the diagonal.

Uniform differentiability and (21) imply that, for all sufficiently small $h > 0$,

$$|v_h(z) - z| \le Ch, \qquad h \le 1 - |v_h(z)|^2 \le 3h,$$

uniformly in $z$. Since $|w - v_h(z)| \ge 1 - |v_h(z)| \ge h/2$, the triangle inequality gives

$$|w - z| \le |w - v_h(z)| + Ch \le (1 + 2C)|w - v_h(z)|.$$

Consequently

$$h^{-1} p_{v_h(z)}(w) |w - z|^2 \le 3(1 + 2C)^2. \tag{23}$$

Suppose $f \ge \delta > 0$ and $|f(z) - f(w)| \le L|w - z|$. Taylor's integral remainder for $s \log s - s$ yields

$$0 \le B(f(z), f(w)) \le \frac{L^2}{2\delta} |w - z|^2.$$

Combine this bound with (23) and the positive lower bound of $p_q$ on the circle. The functions

$$\frac{p_{v_h(z)}(w)}{h \, p_q(w)} B(f(z), f(w))$$

are uniformly bounded on $\mathbb{T}^2$, with value zero when $z = w$. Their pointwise limit is the singular integrand in (6), with the stated zero diagonal convention. Dominated convergence proves its integrability and the equality of its integral with the limit in (20). Apply this to $f_t$ using the positivity and Lipschitz bounds from the preceding section. This proves the exact production formula.

The drift contributes no separate square-gradient term to this formula. Its effect is retained in $f_t$ and the invariant law; (21) shows that the off-diagonal jump rate is the Cauchy rate. No diffusion chain rule or symmetry assumption enters this argument.

## 8 A nonsharp entropy inequality and decay from initial time

For $w \in \mathbb{T}$,

$$p_q(w) \le \frac{1+|q|}{1-|q|}, \qquad |w-z| \le 2.$$

Thus away from the diagonal $2/(|w-z|^2 p_q(w)) \ge c_q$. Since $B(f(z), f(z)) = 0$, the product with $B$ satisfies the same inequality everywhere under the zero convention. For a normalized positive Lipschitz $f$ bounded away from zero, Fubini gives

$$\begin{aligned}\int_{\mathbb{T}^2} B(f(z), f(w))\, d\nu_q(z) d\nu_q(w) &= \int_{\mathbb{T}} f \log f \, d\nu_q - \int_{\mathbb{T}} \log f \, d\nu_q \\ &\ge \operatorname{Ent}_{\nu_q}(f).\end{aligned}$$

The last step is Jensen's inequality for the logarithm and $\int f \, d\nu_q = 1$. We have proved the modified logarithmic-Sobolev inequality in the exact form

$$c_q \operatorname{Ent}_{\nu_q}(f) \le \int_{\mathbb{T}^2} \frac{2B(f(z), f(w))}{|w-z|^2 p_q(w)} \, \nu_q(dz)\nu_q(dw). \tag{24}$$

This is a jump-Bregman inequality on the displayed positive Lipschitz class, rather than a sharp quadratic Dirichlet-form logarithmic-Sobolev assertion.

For the actual $f_t$, the production formula implies $E'(t) \le -c_q E(t)$ for $t > 0$. Multiplying by $e^{c_q t}$ and integrating yields

$$E(t) \le e^{-c_q(t-s)} E(s), \qquad 0 < s \le t. \tag{25}$$

To reach initial time, we also need entropy contraction without any initial smoothness.

If $D(\mu\|\nu_q) < \infty$, write $\mu = f\nu_q$. For $h > 0$ put $R_h(z,w) = p_{\phi_h(z)}(w)/p_q(w)$. Invariance says $\int R_h(z,w)\nu_q(dz) = 1$ for $\nu_q$-almost every $w$; the row integral is also one. The output density is $f_h(w) = \int R_h(z,w) f(z) \nu_q(dz)$. Apply Jensen's inequality pointwise for the nonnegative convex function $\Psi(a) = a\log a - a + 1$, then integrate:

$$\int \Psi(f_h)\, d\nu_q \le \iint R_h(z,w)\Psi(f(z))\, d\nu_q(z) d\nu_q(w) = \int \Psi(f)\, d\nu_q.$$

Tonelli justifies the extended nonnegative integrals. This is $E(h) \le D(\mu\|\nu_q)$. Insert it in (25) and let $s \downarrow 0$ to obtain (7) when the initial entropy is finite. If it is infinite the same extended-value inequality is immediate; at $t = 0$ it is equality. Positive-time finiteness has already been proved directly from (17), even for singular initial laws.

For any fixed $s > 0$, (25) also proves $E(t) \to 0$ for every initial probability law as $t \to \infty$. If a probability measure satisfies $\mu K_h = \mu$ for one $h > 0$, it has finite entropy by positive-time smoothing. Equation (7) then forces $E \le e^{-c_q h} E$, so $E = 0$. Strict convexity of $\Psi$ gives $\mu = \nu_q$. This proves the uniqueness assertion without a path-space construction or a reversibility premise.

## 9 Exact finite-core drift corrections

The preceding entropy theorem uses the actual invariant law $\nu_q$. We now give finite polynomial tests that explain why its proof cannot be replaced by a diffusion calculation under Haar measure. All operators in this section act on the algebra of finite trigonometric polynomials; their products and derivatives stay in that algebra. Write $e_n(\theta) = e^{in\theta}$, $L_0 e_n = -|n| e_n$, and, for an integer $h \ge 0$, put

$$B_h f = -\sin(h\theta) f', \qquad L_{a,h} = L_0 + aB_h, \quad a \in \mathbb{R}.$$

Here $h$ denotes a harmonic index, not a time step; the generator already constructed is $L_{r,1}$ with the amplitude $r$ from (1). No transition family for arbitrary $h$ is being asserted. For a real-preserving operator $L$ define

$$\Gamma_L(f,g) = \tfrac{1}{2}\{L(\bar f g) - \overline{Lf}\, g - \bar f Lg\},$$
$$\Gamma_{2,L}(f,g) = \tfrac{1}{2}\{L\Gamma_L(f,g) - \Gamma_L(Lf,g) - \Gamma_L(f,Lg)\}.$$

The conjugate in the first argument is essential. For $L_0$ let $\Gamma = \Gamma_{L_0}$ and $\Gamma_2 = \Gamma_{2,L_0}$. Directly on modes,

$$\Gamma(e_n,e_m) = K(n,m)e_{m-n}, \quad \Gamma_2(e_n,e_m) = K(n,m)^2 e_{m-n}, \quad K(n,m) = \frac{|n|+|m|-|m-n|}{2}. \tag{26}$$

Indeed the first formula follows by applying $L_0$ to each of the three terms, and the second multiplies the first coefficient by $(|n|+|m|-|m-n|)/2$ again. In particular $K$ is zero when its indices have opposite signs and is $\min(|n|,|m|)$ when they have the same sign; either description gives zero at a zero index.

Every $B_h$ commutes with conjugation and satisfies the product rule. Consequently

$$\Gamma_{L_{a,h}} = \Gamma, \qquad \Gamma_{2,L_{a,h}} - \Gamma_2 = \frac{a}{2}\mathcal{C}_h, \quad \mathcal{C}_h(f,g) = B_h\Gamma(f,g) - \Gamma(B_h f,g) - \Gamma(f,B_h g). \tag{27}$$

The first assertion follows by cancelling the three derivation terms. Substitute it into the definition of the second form; the remaining $L_0$ terms give $\Gamma_2$ and the remaining drift terms give the displayed commutator. This proves the identities for every finite polynomial, without a domain limit.

**Proposition 9.1.** *For every real amplitude $a$ and all finite $f, g$,*

$$\Gamma_{2,L_{a,1}}(f,g) = \Gamma_2(f,g) + \frac{a}{2}\cos\theta\,\Gamma(f,g).$$

*Proof.* The mode action is $B_1 e_n = (n/2)(e_{n-1} - e_{n+1})$. For the coefficient of $e_{m-n-1}$ in $\mathcal{C}_1(e_n,e_m)$, (26) gives one half of

$$(m-n)K(n,m) + nK(n+1,m) - mK(n,m-1) = K(n,m). \tag{28}$$

Here is a complete sign check. If $n \ge 0, m \ge 1$ and $m \ge n+1$, the three kernels on the left are $n, n+1, n$; the left side is $n$. If instead $1 \le m \le n$, they are $m, m, m-1$, and the left side is $m$. If $n \ge 0, m \le 0$, all three kernels vanish. They also vanish if $n \le -1, m \ge 1$. Finally if $n \le -1, m \le 0$, set $N = -m \ge 0, M = -n \ge 1$. By symmetry and simultaneous sign reversal, the expression becomes

$$(M-N)K(N,M) + NK(N+1,M) - MK(N,M-1),$$

which is the already proved positive case. These cases cover all integer pairs. The coefficient at $e_{m-n+1}$ is also $K(n,m)/2$: its identity is obtained from (28) by $(n,m) \mapsto (-n,-m)$ and $K(-n,-m) = K(n,m)$. There are no other frequencies. Thus $\mathcal{C}_1(e_n,e_m) = \cos\theta\,\Gamma(e_n,e_m)$. Conjugate linearity in $f$ and linearity in $g$ extend this equality to their finite sums. Equation (27) finishes. □

## 10 The single-harmonic scalar correction is exceptional

**Theorem 10.1.** *Among integer harmonics $h \ge 0$, a function $b(\theta)$ such that*

$$\Gamma_{2,L_{1,h}}(f,f)(\theta) - \Gamma_2(f,f)(\theta) = b(\theta)\Gamma(f,f)(\theta)$$

*for every real trigonometric polynomial $f$ and every $\theta$ exists exactly when $h = 0$ or $h = 1$. The same classification holds if all complex polynomials are required.*

*Proof.* For $h = 0$, $B_h = 0$ and $b = 0$ works. For $h = 1$ take $b = \cos\theta/2$ by Proposition 9.1. For the converse, put $p_n = e_n + e_{-n} = 2\cos(n\theta)$, $n > 0$. Equation (26) gives $\Gamma(p_n, p_n)(0) = 2n$. At $\theta = 0$ one has $B_h\Gamma(p_n, p_n) = 0$, since $\sin 0 = 0$. The two remaining real diagonal commutator terms agree. Also

$$B_h p_n = \frac{n}{2}(e_{n-h} - e_{n+h} - e_{-n-h} + e_{-n+h}).$$

For any integer $j$, the sum $K(j, n) + K(j, -n) = \min(|j|, n)$. Hence direct substitution gives

$$\Gamma(B_h p_n, p_n)(0) = n\{\min(|n-h|, n) - n\},$$
$$[\Gamma_{2,L_{1,h}}(p_n, p_n) - \Gamma_2(p_n, p_n)](0) = n\{n - \min(|n-h|, n)\}.$$

If $h \geq 2$, the value is zero for $n = 1$, forcing $b(0) = 0$. For $n = h$ the value is $h^2 > 0$, contradicting the same factor. These are genuine real witnesses, so allowing complex polynomials cannot repair the failure. □

For comparison, the complex diagonal has the full frequency formula

$$\Gamma_{2,L_{1,h}}(e_n, e_n) - \Gamma_2(e_n, e_n) = \frac{n\min(n, h)}{4}(e_h + e_{-h}), \qquad n, h \geq 0. \tag{29}$$

To verify it, $\Gamma(e_n, e_n) = n$ is constant and $B_h n = 0$. The action $B_h e_n = (n/2)(e_{n-h} - e_{n+h})$ in the other two commutator terms leaves the difference

$$K(n + h, n) - K(n - h, n) = \min(n, h).$$

For $n \geq h$ its two terms are $n, n - h$; for $0 \leq n \leq h$ they are $n, 0$. Multiplication by $n/4$ proves (29). At zero, the correction divided by $\Gamma(e_n, e_n) = n$ is $\min(n, h)/2$ for $n > 0$, again incompatible with a common scalar for $h \geq 2$. The stronger real-witness calculation above is needed to exclude an artefact of complex tests. For a fixed nonzero amplitude $a$, all corrections scale by $a$, so the same classification follows. Zero amplitude gives the drift-free identity for every $h$. This last amplitude observation is a direct deduction, rather than a separate named formal theorem. No classification of mixtures of harmonics or arbitrary potentials is inferred from a classification of one harmonic at a time.

# 11 Two independent obstructions to diffusion shortcuts

## 11.1 The actual Haar energy can be negative

Let $L = L_{1,1}$. The real positive polynomial $f = 2 + \cos\theta$ satisfies

$$Lf = -\cos\theta + \sin^2\theta, \qquad \int Lf\, dm = \tfrac{1}{2}.$$

Thus Haar measure is not stationary even on this finite core. Using $\int \cos\theta\, dm = \int \cos^3\theta\, dm = 0$ and $\int \cos^2\theta\, dm = \int \sin^2\theta\, dm = 1/2$, one gets

$$\int f\, dm = 2, \qquad \int f^2\, dm = \tfrac{9}{2}, \qquad -\int fLf\, dm = -\tfrac{1}{2}.$$

The last equality follows from $\int(2 + \cos\theta)(-\cos\theta + \sin^2\theta)\, dm = -1/2 + 1 = 1/2$. Its variance is $1/2 > 0$, so there is no constant $C \geq 0$ for which

$$\mathrm{Var}_m(f) \leq C(-\int fLf\, dm)$$

holds for every real finite polynomial. In this counterexample the right side is nonpositive. This is a failure of the stated uncentered Haar-energy inequality: centering the test before applying the nonstationary energy changes that energy. It is not a refutation of all possible centered or invariant-law Poincaré inequalities. Amplitude one is included in the preceding semigroup: it corresponds to $q = \sqrt{2} - 1$, for which the correct reference law is the nonuniform $\nu_q$.

### 11.2 A cubic chain-rule remainder survives every sine amplitude

A diffusion operator would satisfy $L(g^3) = 3g^2Lg + 6g\Gamma_L(g,g)$ for real $g$. Take $g = \cos\theta$. From $g^2 = (1+\cos 2\theta)/2$ and $g^3 = (3\cos\theta + \cos 3\theta)/4$,

$$L_0 g(0) = -1, \quad L_0(g^2)(0) = -1, \quad L_0(g^3)(0) = -\tfrac{3}{2}, \quad \Gamma(g,g)(0) = \tfrac{1}{2}.$$

The diffusion right side would be $-3 + 3 = 0$, which differs from $-3/2$. More structurally define

$$\mathcal{D}_L(g) = L(g^3) - 3gL(g^2) + 3g^2Lg.$$

For a derivation $B$, the product rule gives $B(g^2) = 2gBg$ and $B(g^3) = 3g^2Bg$; its three contributions are $3 - 6 + 3 = 0$. Thus $\mathcal{D}_{L_0 + aB_1} = \mathcal{D}_{L_0}$ for every real $a$, while the value on this cosine at zero is $-3/2$. The failure is intrinsic to the nonlocal Cauchy part and is not repaired by its first-harmonic scalar $\Gamma_2$ correction. All evaluations use actual smooth real polynomials.

These tests leave the earlier positive entropy theorem intact: that proof uses the true stationary law and a jump-Bregman identity, not a diffusion chain rule. They also do not disprove a particular nonlocal curvature inequality. Spener–Weber–Zacher [3] study a distinct question, finite-dimensional curvature-dimension bounds for the fractional Laplacian on Euclidean space. Their primary introduction and definitions place nonlocal $\Gamma$ calculus in its established setting; their Euclidean theorem is not being transferred to this circle model.

## 12 Forward law duality and point-mass norm gaps

Let $E = C(\mathbb{T};\mathbb{C})$ with its supremum norm. Its continuous complex dual $E'$ carries the operator norm. Define the transpose semigroup by

$$P_t'\varphi(f) = \varphi(P_t f), \qquad \varphi \in E', \quad t \geq 0.$$

It is contractive, and for every fixed $f \in E$ the scalar function $t \mapsto P_t'\varphi(f)$ is continuous. Both statements follow directly from contractivity and strong continuity of $P_t$ on $E$. For a probability law $\mu$, its functional $\varphi_\mu(f) = \int f\,d\mu$ has norm one: the integral bound gives norm at most one, and testing the constant function one gives equality. The kernel pairing identifies the *actual* forward law:

$$P_t'\varphi_\mu(f) = \int P_t f\,d\mu = \int f\,d(\mu K_t).$$

For every $f \in D(A)$, the full generator identity (5), followed by this bounded functional, gives

$$\int f\,d\mu_t - \int f\,d\mu = \int_0^t \int Af\,d\mu_s\,ds. \tag{30}$$

The integrand is continuous in $s$, so its scalar derivative at every positive time is $\int Af\,d\mu_t$. This weak evolution uses the full closed generator domain, not just a Fourier polynomial test class.

*Source correspondence:* `CauchyPoissonForwardDual.actual_law_duality`, `weak_forward_integrated`, `weak_forward_derivative`, `dualOrbit_weak_continuous`.

The corresponding norm assertion fails for every admissible drift. Let $\delta_x$ also denote evaluation at $x \in \mathbb{T}$, and take $f_n(z) = z^n$, whose norm is one. Harmonic extension gives $P_t f_n(x) = \phi_t(x)^n$. Since $|\phi_t(x)| < 1$ for $t > 0$,

$$\|P_t'\delta_x - \delta_x\| \geq |\phi_t(x)^n - x^n| \geq 1 - |\phi_t(x)|^n \longrightarrow 1 \quad (n \to \infty).$$

Thus the dual norm gap is at least one at every positive time, even as $t \downarrow 0$. This is compatible with convergence against each fixed continuous test: the detecting powers depend on the norm test.

*Source correspondence:* `CauchyPoissonDualObstruction.dirac_dual_norm_gap`, `dirac_dual_not_strong_right`.

There is a generator-domain obstruction as well. Define the Banach transpose $A'$ by $\varphi \in D(A')$ if there exists $\psi \in E'$ such that $\varphi(Af) = \psi(f)$ for every $f \in D(A)$. Density of $D(A)$ makes $\psi$ unique. At $q = 0$, $Af_n = -nf_n$. If evaluation $\delta_x$ had a bounded transposed value $\psi$, then $|\psi(f_n)| = n$, contradicting $\|f_n\| = 1$ and boundedness of $\psi$. Hence $\delta_x \notin D(A')$ at zero drift. This concerns the continuous-function dual; it is not a characterization of a forward density generator on some other normed space.

*Source correspondence:* `CauchyPoissonDualObstruction.zero_dirac_no_adjoint_value`.

## 13 Time averages and the transposed generator

For $T > 0$ define the bounded orbit average on $E$ and its dual action by

$$B_T f = \frac{1}{T}\int_0^T P_s f \, ds, \qquad \overline{\varphi}_T(f) = \varphi(B_T f). \tag{31}$$

The integral here is an ordinary Bochner integral in $E$, where the orbit is strongly continuous. Thus this definition does not presuppose a norm-Bochner integral of the generally discontinuous dual orbit. It gives $\|B_T\| \leq 1$ and $\|\overline{\varphi}_T\| \leq \|\varphi\|$. For a probability law functional it reads, on each continuous test,

$$\overline{\varphi}_{\mu,T}(f) = \frac{1}{T}\int_0^T \int f \, d(\mu K_s)\, ds.$$

Only this functional and scalar identity are needed here; no new measure-valued average construction is assumed from the formal source.

**Proposition 13.1** (Exact regularity supplied by averaging)**.** *For every $\varphi \in E'$ and $T > 0$,*

$$\overline{\varphi}_T \in D(A'), \qquad A'\overline{\varphi}_T = \frac{P_T'\varphi - \varphi}{T}, \tag{32}$$

$$\|P_h'\overline{\varphi}_T - \overline{\varphi}_T\| \leq \frac{2h}{T}\|\varphi\| \qquad (h \geq 0). \tag{33}$$

*Also $\overline{\varphi}_T(f) \to \varphi(f)$ as $T \downarrow 0$ for every fixed $f \in E$.*

*Proof.* For $f \in D(A)$, integrate the generator orbit identity to get $B_T A f = (P_T f - f)/T$. Applying $\varphi$ proves (32); its right side is a bounded functional of norm at most $2\|\varphi\|/T$. The semigroup property and bounded integration give

$$\begin{aligned} B_T P_h f - B_T f &= \frac{1}{T}\left(\int_h^{T+h} P_s f \, ds - \int_0^T P_s f \, ds\right) \\ &= \frac{1}{T}\left(\int_T^{T+h} P_s f \, ds - \int_0^h P_s f \, ds\right). \end{aligned}$$

Each short integral has norm at most $h\|f\|$, proving (33) after applying $\varphi$ and taking the unit-ball supremum. Finally $\|B_T f - f\| \le \sup_{0\le s\le T}\|P_s f - f\| \to 0$ by strong continuity at zero. Applying the bounded functional proves the last claim. □

*Source correspondence:* `CauchyPoissonDualAverage.average_adjoint_pairing`, `average_actual_law`, `average_weak_limit`, `average_strong_bound`, `average_strong_right`.

The bounded transposed value in (32) is not an automatic norm derivative of this dual orbit. The following quantitative test is a *human deduction* from the zero-drift Fourier formulas, not a named formal theorem.

**Proposition 13.2** (A continuous averaged orbit with no norm derivative)**.** *Take $q = 0$, $x \in \mathbb{T}$, $T > 0$ and $\overline{\delta}_{x,T} = \overline{\varphi}_T$ for $\varphi = \delta_x$. Its dual orbit is norm continuous by* (33)*, but has no norm right derivative at zero. In fact, with $v_T = (P_T'\delta_x - \delta_x)/T$, for every $h > 0$,*

$$\left\| \frac{P_h'\overline{\delta}_{x,T} - \overline{\delta}_{x,T}}{h} - v_T \right\| \ge \frac{1}{T}. \tag{34}$$

*Moreover $\|\overline{\delta}_{x,T} - \delta_x\| \ge 1$ for every $T > 0$, despite its convergence on each fixed test as $T \downarrow 0$.*

*Proof.* At zero drift $P_t f_n = e^{-nt} f_n$. Therefore

$$\overline{\delta}_{x,T}(f_n) = \frac{1 - e^{-nT}}{nT} x^n, \qquad v_T(f_n) = -\frac{1 - e^{-nT}}{T} x^n \quad (n \ge 1).$$

The difference quotient minus $v_T$, tested on $f_n$, is exactly

$$\frac{1 - e^{-nT}}{T}\left(1 - \frac{1 - e^{-nh}}{nh}\right) x^n.$$

For fixed $h, T > 0$ its absolute value tends to $1/T$ as $n \to \infty$. Since $\|f_n\| = 1$, this proves (34). To exclude any other strong derivative candidate, observe that for every fixed $f \in E$ the same interval-shift identity gives

$$\frac{(P_h'\overline{\delta}_{x,T} - \overline{\delta}_{x,T})(f)}{h} = \frac{1}{Th}\left(\int_T^{T+h} (P_s f)(x)\, ds - \int_0^h (P_s f)(x)\, ds\right) \longrightarrow v_T(f)$$

by continuity of the scalar orbit. A norm derivative would have to be this weak limit, which (34) excludes. Finally the first Fourier-average coefficient above tends to zero as $n \to \infty$ for fixed $T$, whereas $|\delta_x(f_n)| = 1$. Taking this limit in the unit-test norm bound proves the final gap. □

These distinctions belong to classical adjoint-semigroup theory. Van Neerven–de Pagter [1], introduction and Section 1, describe the norm-continuity subspace of an adjoint semigroup and weak-star orbit integrals in the transposed-generator domain. Our construction uses the explicit circle kernel and its Fourier tests to display these boundaries, without a priority claim. The weak law evolution, positive-time entropy theorem, bounded transposed value, norm-continuous averaged orbit and norm differentiability are separate assertions; the failure of the last does not contradict the others.

## 14 The full drift-free Haar-$L^2$ graph and unsmoothed core

The drift-free benchmark has a complete $L^2$ domain theorem. It is useful to state it separately from the drifted continuous-function generator and the forward-law dual domains above. Identify $\mathbb{R}/\mathbb{Z}$ with the angular circle by $\theta = 2\pi x$, and use normalized Haar measure $dx = d\theta/(2\pi)$. Let $E = L^2(\mathbb{R}/\mathbb{Z}; \mathbb{C})$, with its actual Fourier orthonormal basis $e_n(x) = e^{2\pi i n x}$, $n \in \mathbb{Z}$. Write $\widehat{f}(n) = \langle e_n, f\rangle$; Parseval gives $\|f\|_2^2 = \sum_n |\widehat{f}(n)|^2$.

More generally, prescribe any rates $a_n \geq 0$, without a symmetry, growth or probability hypothesis. The diagonal maps

$$\widehat{T_t f}(n) = e^{-a_n t}\widehat{f}(n), \qquad t \geq 0,$$

are well-defined contractions of this complex $L^2$ space by Parseval. Coordinatewise multiplication proves $T_0 = I$ and $T_s T_t = T_{s+t}$. For $t \to s \geq 0$, each coefficient difference tends to zero and its squared absolute value is bounded by $4|\widehat{f}(n)|^2$. Dominated convergence of this summable sequence proves strong continuity of $t \mapsto T_t f$.

**Theorem 14.1.** *The right norm generator $A_a$ of $T$ has exactly the domain*

$$\operatorname{Dom} A_a = \left\{ f \in E : \sum_{n\in\mathbb{Z}} a_n^2 |\widehat{f}(n)|^2 < \infty \right\}, \qquad \widehat{A_a f}(n) = -a_n \widehat{f}(n).$$

*Its graph is closed, and finite trigonometric polynomials form a graph core without time smoothing.*

*Proof.* For each $b \geq 0$ and $t > 0$, the elementary exponential inequalities $1 - bt \leq e^{-bt} \leq 1$ give

$$\left| \frac{e^{-bt} - 1}{t} \right| \leq b, \qquad \frac{e^{-bt} - 1}{t} \longrightarrow -b.$$

If $t^{-1}(T_t f - f) \to g$ in $E$, every continuous Fourier coordinate functional passes this limit and gives $\widehat{g}(n) = -a_n \widehat{f}(n)$. Parseval for $g$ then forces the stated weighted square summability. Conversely, if that summability holds, the Fourier basis defines the unique $g \in E$ with those coefficients. The coefficient difference between the quotient and $g$ tends to zero and is bounded by $2a_n|\widehat{f}(n)|$. Dominated convergence of its square proves convergence in $E$. This establishes both directions of the exact right-generator graph.

For closedness, if $f_j \to f$, $g_j \to g$ in $E$ with $\widehat{g}_j(n) = -a_n \widehat{f}_j(n)$, pass every coefficient identity to the limit. The resulting $g$ has the required coefficients, so the pair is in the same graph. For any graph pair $(f, A_a f)$, truncate both Fourier series to $|n| \leq N$. Their tails tend to zero in their respective $L^2$ norms by Parseval, and each truncated pair remains in the same graph. Thus these finite inputs are dense in the graph norm. They are also dense in $E$, proving that this closed generator is densely defined. No change of graph or native derivative value occurred in either direction. □

The formal graph-core theorem is stated as the closure of all finite Fourier graph pairs, with truncations over finite subsets of $\mathbb{Z}$. The symmetric sequence $\{-N, \ldots, N\}$ here is a cofinal human choice of those same truncations.

`FractionalFourierGraphCore`:
`graph_domain_iff`, `graph_closed`, `finite_graph_closure`.
`FractionalFourierSemigroup`:
`strong_continuity`, `generator_iff`.

For $\gamma > 0$, take $a_n = |n|^\gamma$, with $a_0 = 0$. The exact fractional domain and multiplier are therefore

$$\operatorname{Dom} A_\gamma = \left\{ f \in E : \sum_n |n|^{2\gamma} |\widehat{f}(n)|^2 < \infty \right\}, \qquad A_\gamma = -(-\partial_\theta^2)^{\gamma/2}.$$

The angular coordinate is part of this equality: the unit-period $x$-Laplacian has eigenvalues $(2\pi n)^2$, so its corresponding fractional operator has an additional factor $(2\pi)^\gamma$. We restrict to $\gamma > 0$ here to avoid any ambiguity at the zero frequency. All these diagonal families are $L^2$ contractions; this argument alone makes no positivity or Markov assertion for an arbitrary rate sequence or arbitrary $\gamma$. Roncal–Stinga's primary Fourier definition and opening Section 5 give the classical weighted-$L^2$ Sobolev domain with angular periods $2\pi$ [4]. No new fractional-domain principle is claimed.
`FractionalFourierSemigroup`:
`fractional_generator_domain`, `fractional_time_core`,
`actual_fourier_evolution`.

For $\gamma = 1$ this is exactly the Haar-$L^2$ extension of the zero-drift Cauchy kernel already constructed in this paper. Indeed the disk flow becomes $z \mapsto e^{-t} z$, and for $t > 0$ its angular transition density relative to normalized Haar is

$$p_t(\theta) = \frac{1 - e^{-2t}}{1 - 2e^{-t}\cos\theta + e^{-2t}}.$$

The two geometric series give its absolutely convergent expansion $p_t(\theta) = \sum_{n\in\mathbb{Z}} e^{-|n|t} e^{in\theta}$. Hence convolution acts on every trigonometric polynomial by the stated diagonal multiplier. Haar invariance and Jensen's inequality make this actual kernel an $L^2$ contraction, so density extends the agreement to all of $E$. This kernel-to-diagonal identification is a human consequence of the existing zero-drift formulas.

The result supplies the full unsmoothed Haar-$L^2$ graph core in this drift-free case. It does not identify a maximal forward-density domain, a pointwise Gamma extension, or the unsmoothed graph core of the nonsymmetric sine-drift generator. The Fourier modes of that drifted generator are coupled, and its stationary measure is the wrapped Cauchy law rather than Haar when the drift is nonzero. The zero-drift domain theorem cannot silently substitute for those different domains.

## 15 Classical context and exact verification scope

The Poisson kernel and harmonic-composition construction are classical. The pole equation $\dot{a} = -a + (r/2)(1 - a^2)$ is also the constant-forcing, zero-frequency, unit-noise specialization of the Cauchy-driven phase equation in Tanaka [2], equations (9) and (17). Its relation to a known invariant family must therefore be acknowledged; we do not present the rational pole flow as a new general mechanism. The generator/resolvent arguments are standard contraction semigroup theory. The complete kernel-to-forward-entropy exposition and its precise formal domains are the focus here. Priority for this explicit combined formulation has not been established.

All mapped statements use the frozen verified baseline

`bab688bb6140cd6316c80680095a914ea0d346db.`

The accompanying theorem map covers the actual Markov kernels, Feller semigroup, closed generator, smoothed core, Laplace resolvent, forward density, stationary Bregman identity, singular limit, modified entropy bound and finite nonsymmetry witness. The additional finite-core map records the actual sine correction, single-harmonic classification, real cosine obstructions, Haar-energy failure and cubic defect. The one-dimensional formulas are the coordinate-axis specialization of the formal

three-coordinate finite Fourier algebra, with all other coordinates zero. The human proof groups the same ingredients into harmonic and Banach-space arguments; intermediate regroupings are not claimed as new named Lean theorems. The added forward-dual and averaging map retains the Banach transpose domain and weak-star topology. The explicit zero-drift failure of norm differentiability and the averaged Dirac norm gap are labelled human deductions. The nine retained exact finite-core checks remain; the new functional/domain limits are verified by their complete analytic proofs rather than numerical sampling.

The results concern the displayed circle kernels, the continuous-function generator and the additional drift-free Haar-$L^2$ graph theorem. A trajectory law, maximal forward-density generator domain, unsmoothed Fourier graph core for the drifted operator, sharp entropy constant or diffusion curvature estimate is not asserted here. In particular an invariant wrapped Cauchy law cannot justify replacing the actual nonsymmetric operator by a symmetric diffusion.

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